



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)



KJSSE/AIDS/SYBTECH /SEM-III/FDS/2026-27

Experiment No. : 3

Title: Measuring Central Tendency and variability of the Data

Batch: A-1(AIDS) Roll No.: 16014225013**Experiment No.: 3**

Aim: To compute and analyze the measures of central tendency (Mean, Median, Mode) and variability (Range, Variance, Standard Deviation, Interquartile Range) for single and multi-attribute datasets using statistical analysis in Python.

Resources needed: Resources needed: Python 3.x environment (Jupyter Notebook / Google Colab etc.), any statistical analysis tool

Theory

Mathematically, central tendency means measuring the center or distribution of location of values of a data set. It gives an idea of the average value of the data in the data set and also an indication of how widely the values are spread in the data set. That in turn helps in evaluating the chances of a new input fitting into the existing data set and hence probability of success.

Measures of Central Tendency:

There are three main measures of central tendency:

- **Mean:** It is the average value of the data, which is a division of the sum of the values by the number of values.
- **Median:** It is the middle value in a distribution when the values are arranged in ascending or descending order.
- **Mode:** It is the most commonly occurring value in a distribution.

Measures of Variability:

Variability describes how far apart data points lie from each other and from the center of a distribution. Along with measures of central tendency, measures of variability give you descriptive statistics that summarize your data. Variability is also referred to as spread, scatter, or dispersion. It is most commonly measured with the following:

- **Range:** The difference between the highest and lowest values.
- **Interquartile Range (IQR):** The range of the middle half of a distribution.
- **Standard Deviation:** Average distance from the mean.
- **Variance:** Average of squared distances from the mean.
- **Coefficient of Variation:** Ratio of the standard deviation to the mean.

Activity:

1. Analyze the data to find out frequency, mean, mode, median, standard deviation, variance, and interquartile range of the data.
2. Write a programming code with a procedure to compute mean, mode, median, variance, standard deviation, and interquartile range of a given sample data **without using in-built functions**.
3. Compute and analyze the grouped data to find out frequency, mean, mode, median, standard deviation, variance, and interquartile range of data.

Results: (program with output)

1. Analyze the data to find out frequency, mean, mode, median, standard deviation, variance, and interquartile range of the data.

Program

```
import pandas as pd
import numpy as np

df = pd.read_csv('delhi-weather-aqi-2025.csv')
print(df.shape)
df.head()

# ----- (a) Single attribute: temp_c -----
temp = df['temp_c']

mean_val = temp.mean()
median_val = temp.median()
mode_val = temp.mode()[0]
range_val = temp.max() - temp.min()
variance_val = temp.var()
std_val = temp.std()
Q1 = temp.quantile(0.25)
Q3 = temp.quantile(0.75)
IQR_val = Q3 - Q1
cov_val = std_val / mean_val
print("Frequency (top 5 values):")
print(temp.value_counts().head())

print(f"Mean                : {mean_val:.2f}")
print(f"Median                : {median_val:.2f}")
print(f"Mode                  : {mode_val:.2f}")
print(f"Range                 : {range_val:.2f}")
print(f"Variance              : {variance_val:.2f}")
print(f"Standard Deviation    : {std_val:.2f}")
print(f"Q1, Q3                : {Q1:.2f}, {Q3:.2f}")
print(f"IQR                  : {IQR_val:.2f}")
print(f"Coefficient of Var    : {cov_val:.2f}")
```

Output

```
... (52560, 16)
Frequency (top 5 values):
temp_c
27.5    570
27.0    542
28.0    525
26.5    508
28.5    476
Name: count, dtype: int64
Mean      : 25.02
Median    : 26.50
Mode      : 27.50
Range     : 38.10
Variance  : 58.04
Standard Deviation : 7.62
Q1, Q3    : 19.50, 30.50
IQR       : 11.00
Coefficient of Var : 0.30
```

2. Write a programming code with a procedure to compute mean, mode, median, variance, standard deviation, and interquartile range of a given sample data **without using in-built functions**.

Program

```
sample_df = df[df['location'] == 'Anand Vihar'].head(10)
data = sample_df['temp_c'].tolist()
n = len(data)
print("Sample data:", data)

total = 0
for x in data:
    total = total + x
mean_manual = total / n
print("Manual Mean:", mean_manual)

sorted_data = sorted(data)
if n % 2 == 0:
    median_manual = (sorted_data[n // 2 - 1] + sorted_data[n // 2]) / 2
else:
    median_manual = sorted_data[n // 2]
print("Manual Median:", median_manual)
```

```
freq = {}  
for x in data:  
    freq[x] = freq.get(x, 0) + 1  
max_count = max(freq.values())  
modes = [value for value, count in freq.items() if count == max_count]  
print("Frequency table:", freq)  
print("Manual Mode:", modes)
```

```
range_manual = max(data) - min(data)  
  
squared_diff_sum = 0  
for x in data:  
    squared_diff_sum = squared_diff_sum + (x - mean_manual) ** 2  
variance_manual = squared_diff_sum / n  
std_manual = variance_manual ** 0.5  
  
print("Manual Range:", range_manual)  
print("Manual Variance:", variance_manual)  
print("Manual Standard Deviation:", std_manual)
```

```
def manual_median(lst):  
    s = sorted(lst)  
    m = len(s)  
    if m % 2 == 0:  
        return (s[m // 2 - 1] + s[m // 2]) / 2  
    return s[m // 2]  
  
lower_half = sorted_data[: n // 2]  
upper_half = sorted_data[(n + 1) // 2 :]  
Q1_manual = manual_median(lower_half)  
Q3_manual = manual_median(upper_half)  
IQR_manual = Q3_manual - Q1_manual  
print("Manual Q1:", Q1_manual, "| Q3:", Q3_manual, "| IQR:", IQR_manual)
```

output

```
... (52560, 16)
Sample data: [8.1, 7.7, 7.5, 7.8, 7.3, 7.6, 8.1, 8.0, 8.4, 9.4]
Manual Mean: 7.99
Manual Median: 7.9
Frequency table: {8.1: 2, 7.7: 1, 7.5: 1, 7.8: 1, 7.3: 1, 7.6: 1, 8.0: 1, 8.4: 1, 9.4: 1}
Manual Mode: [8.1]
Manual Range: 2.1000000000000005
Manual Variance: 0.3169000000000001
Manual Standard Deviation: 0.562938717801503
Manual Q1: 7.6 | Q3: 8.1 | IQR: 0.5
```

3. Compute and analyze the grouped data to find out frequency, mean, mode, median, standard deviation, variance, and interquartile range of data.

Program

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('delhi-weather-aqi-2025.csv')
```

```
bins = [0, 15, 25, 32, 38, 50]
```

```
labels = ['Cold (<15°C)', 'Mild (15-25°C)', 'Warm (25-32°C)', 'Hot (32-38°C)', 'Heatwave (38°C+)']
```

```
df['temp_category'] = pd.cut(df['temp_c'], bins=bins, labels=labels)
```

```
# Frequency = how many hours fall into each category
```

```
frequency = df['temp_category'].value_counts().sort_index()
```

```
print("Frequency of each temperature category:")
```

```
print(frequency)
```

```
colors = ['#6FA8DC', '#93C47D', '#FFD966', '#E69138', '#CC0000'] # cool blue -> hot red
```

```
plt.figure(figsize=(8, 5))
```

```
bars = plt.bar(frequency.index.astype(str), frequency.values, color=colors, edgecolor='black')
```

```
for bar in bars:
```

```
    height = bar.get_height()
```

```
    plt.text(bar.get_x() + bar.get_width()/2, height + 300, f'{height}',
             ha='center', fontweight='bold')
```

```
plt.title('How Many Hours Fall Into Each Temperature Category (2025)', fontsize=13, fontweight='bold')
```

```
plt.xlabel('Temperature Category')
```

```
plt.ylabel('Number of Hours')
```

```
plt.grid(axis='y', linestyle='--', alpha=0.5)
```

```
plt.tight_layout()
```

```
plt.show()
```

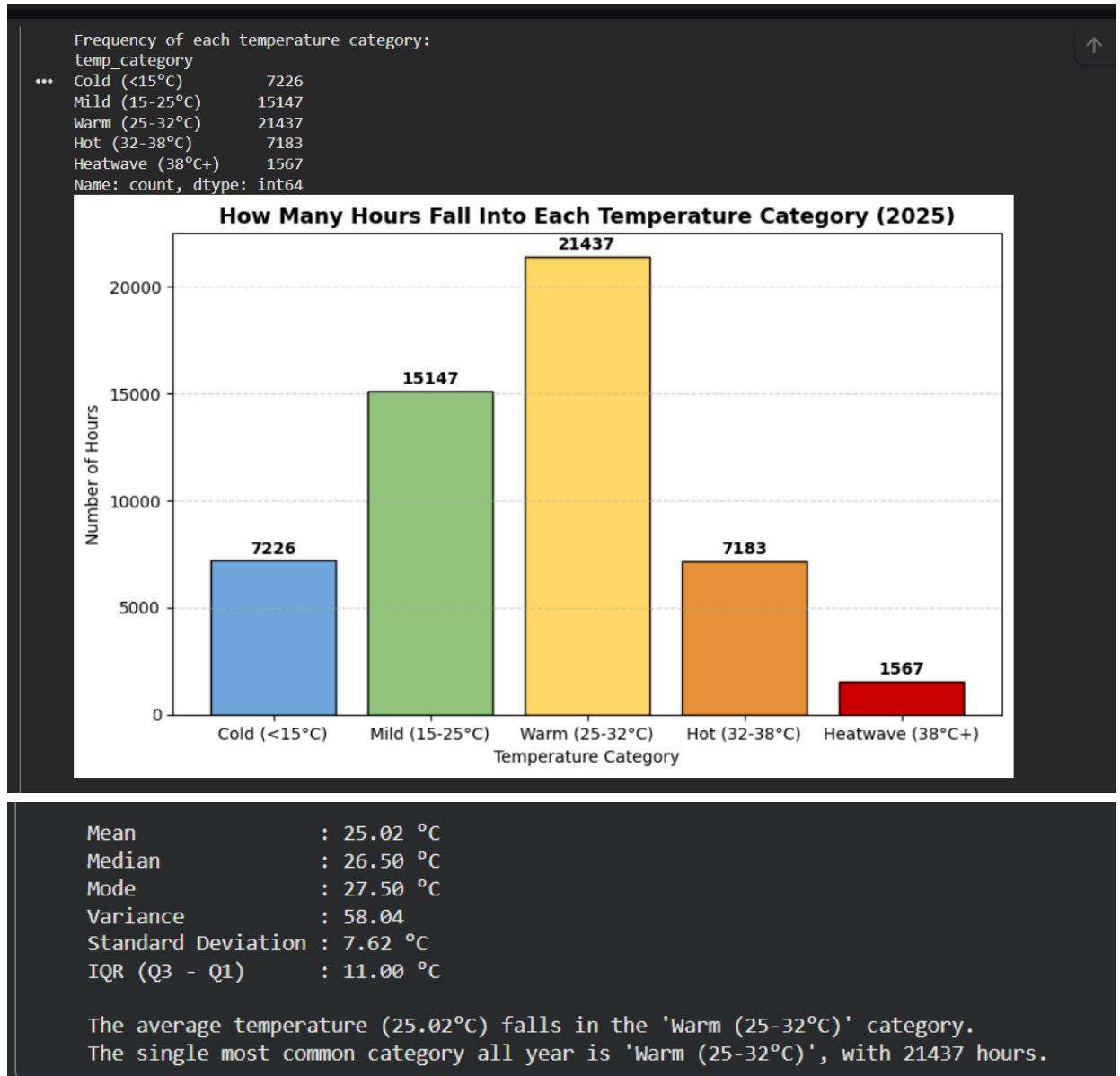
```
mean_val = df['temp_c'].mean()
median_val = df['temp_c'].median()
mode_val = df['temp_c'].mode()[0]
variance_val = df['temp_c'].var()
std_val = df['temp_c'].std()
Q1 = df['temp_c'].quantile(0.25)
Q3 = df['temp_c'].quantile(0.75)
IQR_val = Q3 - Q1
```

```
print(f"\nMean : {mean_val:.2f} °C")
print(f"Median : {median_val:.2f} °C")
print(f"Mode : {mode_val:.2f} °C")
print(f"Variance : {variance_val:.2f}")
print(f"Standard Deviation : {std_val:.2f} °C")
print(f"IQR (Q3 - Q1) : {IQR_val:.2f} °C")
```

```
mean_category = pd.cut([mean_val], bins=bins, labels=labels)[0]
most_common_category = frequency.idxmax()
```

```
print(f"\nThe average temperature ({mean_val:.2f}°C) falls in the '{mean_category}' category.")
print(f"The single most common category all year is '{most_common_category}', with {frequency.max()} hours.")
```

output



Post-Lab Questions:

1. What are the various applications of central tendency and variability of data?

Applications: Central tendency and variability are used almost anywhere data needs to be summarized and understood. In weather and climate studies (like this heatwave dataset), the average temperature tells us what a "typical" day looks like, while the standard deviation and IQR reveal how much temperatures swing — which is exactly what's needed to spot an abnormal heatwave. In business and finance, average sales or median income summarize performance, and the variance or standard deviation of stock returns is used directly as a measure of investment risk. In manufacturing, quality control relies on the mean output staying



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on target with a low standard deviation, which signals a consistent, reliable process. In healthcare, the average recovery time or the spread of blood pressure readings across patients helps doctors judge what counts as "normal" versus something needing attention. And in education, average exam scores combined with the standard deviation tell a teacher not just how well a class performed overall, but how consistent that performance was across students.

2. What are the limitations of Central Tendency and variability of Data?

Limitations: As useful as these measures are, they're far from perfect. The mean, for one, is easily thrown off by outliers a single freak 44°C hour can nudge the whole average upward, even if that hour barely represents what the year was actually like. The median avoids that problem since it only cares about position in the middle, but that's also its weakness, it doesn't tell you anything about how spread out the rest of the values are. The mode gets shaky with continuous data like temperature, where decimal values rarely repeat exactly, so there might be no clear "most common" value at all, or several tied for it. Variance technically captures spread well, but since it's in squared units, it's honestly hard to make sense of on its own that's really why standard deviation exists, to bring it back to something interpretable, though it still isn't immune to outliers either. And maybe the biggest catch is that no single number can fully describe a dataset, two datasets can share the exact same mean and standard deviation and still look completely different when you actually plot them. On top of that, when you're working with grouped data instead of raw values, you're already accepting some loss of accuracy, since every value in a bucket gets treated as if it were the bucket's midpoint.

Course Outcomes: (refer exp list)

Conclusion:

This experiment looked at Delhi's 2025 weather data from three angles — using Python's built-in functions, building the same formulas from scratch, and grouping the data into simple categories. All three approaches agreed with each other, and together they showed that a "typical" hour in Delhi is warm (around 25°C), with real heatwaves being rare, short spikes rather than the norm.

Grade: AA / AB / BB / BC / CC / CD /DD

Signature of faculty in-charge with date

References:



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